

Simple Parts - input file requirements

Simple Parts is a two-part application for uploading client CAD (.dxf or .3dm), flattening 3D geometry to 2D outlines), reviewing part metadata, and downloading the merged result DXF.

For the pipeline to work, the input file needs to follow a set of format and geometrical requirements.

Admitted formats:

.dxf, .3dm, .dwg

Formal/geometrical requirements:

Parts:

Files can contain both 2D curves or 3D geometry.

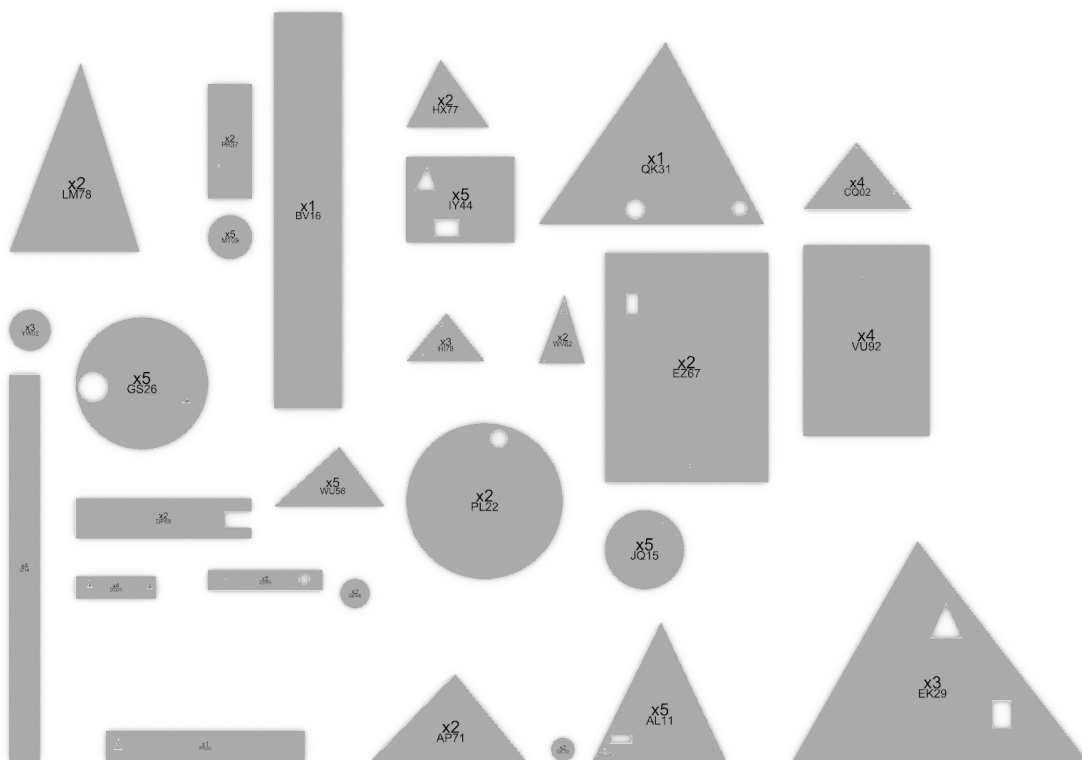
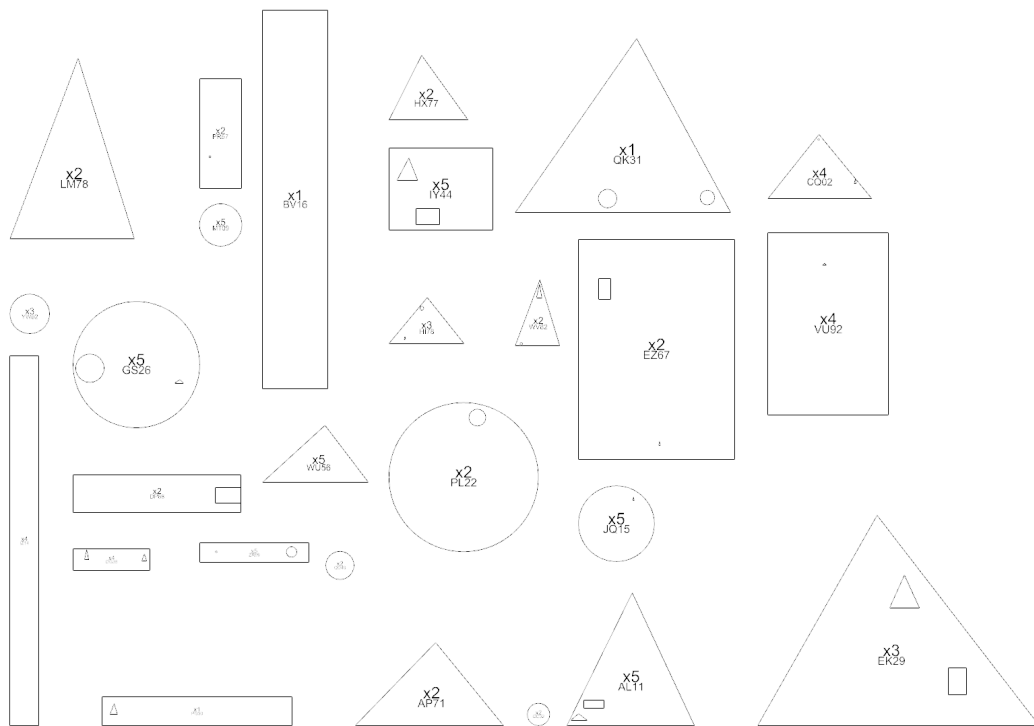
If curves, outlines shall be closed polylines, flattened to $Z=0$, with any necessary cuts made (boundaries within exterior boundaries).

If 3D elements, geometry shall be mesh or brep.

Text/Information (Part name, amount & material):

Nesting works through metadata found on the detected elements. Metadata will be initially read from text elements in the file and can be modified as demanded on the platform, either by clicking on the elements or by asking the chat.

Text shall be placed directly by the part (above/under it) or inside the part. Annotations far away from the corresponding part will yield incorrect results.



Name: Serial Number/Codes will be understood as part names. Redundant text throughout the file will be skipped by default.

Amount: Users can add texts with the following format "7x" to define part amount. In this example, the part will be nested 7 times. If undefined, parts will be set as Amount = 1.

Material: Set directly on the app.

Layers:

Layers are disregarded for computation. We encourage simplistic file structures for faster computation (layer for exterior/interior boundaries, layer for text, etc.)

Other elements:

Annotations, dimensions and any other element outside of CURVES, MESHES, BREPS and TEXT OBJECTS will be disregarded.

Blocks are not recognized: please explode any containing relevant geometry for processing.